Semisupervised Adaptive Ladder Network for Remote Sensing Image Change Detection

Jiao Shi, Tiancheng Wu, A. K. Qin, *Senior Member, IEEE*, Yu Lei, and Gwanggil Jeon

***Abstract*— Nowadays, due to the difficult acquisition of true labels, a semisupervised neural network has shown great potential for change detection (CD) in remote sensing images. However, most of the traditional semisupervised neural network detection frameworks are complex to train and require additional structural analysis, along with a fixed structure, lacking universality. In this article, a semisupervised adaptive ladder network (SSALN) for remote sensing image CD is proposed, which enables dual-input label-incremental architecture searching with a concise and variable structure. First, SSALN is suitable for CD from two remote sensing images of any type with the characteristic of minimal label dependency and automatic network structure adjustment. The network can generate more reliable pseudolabels through continuous iterations to help limited real labels exploit implicit information, identify the most effective network, and form the ascending network structure optimization. Second, the acquisition of pseudolabels is the fusion of semisupervised and unsupervised CD approaches, which ensures the multiperspective information supplement. Multiple CD maps are fused to generate labels for the next iteration, making the predicting more reliable. Finally, both homogenous images and heterogenous images are tested with experiments. Even if the detection object is switched, it can be well adaptive and compatible without manual modification of the network. Experimental results demonstrate that the proposed method can promote the flow of label information through structure searching and self-circulation in the ascending network optimization; thus, it has outstanding performance on tasks of remote sensing image CD.**

***Index Terms*— Adaptive structure, change detection (CD), genetic algorithm, remote sensing image, semisupervised network.**

I. Introduction

REMOTE sensing image change detection (CD) as a research field aims to identify the changes in the same region during different times by comparing and processing a pair of remote sensing images. Along with the development of remote sensing techniques, more types of remote sensing images are gradually diversified, such as synthetic aperture radar (SAR) images, optical images, heterogeneous images, high spatial resolution (VHR) images, multispectral images, and hyperspectral images [1]–[4], producing more abundant remote sensing image information. Nowadays, with more remote sensing image information being able to be fully utilized, the application of remote sensing image CD is common in various kinds of land surface change analysis, such as land cover change, natural disaster assessment, agricultural assessment, urban expansion and evolution, and environmental monitoring [5]–[7].

In the past, many traditional methods have been used for remote sensing image CD. Usually, the difference maps that are preacquired will be further utilized by methods that can evidently analyze the variation characteristics and highlight changes. Most of them are thresholding and clustering methods [8], [9]. Thresholding methods, such as Otsu [10], the Kittler–Illingworth minimum-error thresholding algorithm [11], and the change vector analysis (CVA) [12], [13], are possible to achieve a balance of thresholds between changed pixels and invariant pixels. While they are often sensitive to noise, leading to worse performance in many cases. Meanwhile, in clustering technology applied to remote sensing image CD, CD based on the fuzzy c-means algorithm [14] can reserve more useful information compared with the traditional hard clustering methods. What is more, the principal component analysis (PCA) [15], Markov random fields [16], [17], and genetic algorithm [18], [19] have also been widely used to solve the CD problem. However, the above methods all start from the original pixels, while the detection performance will be greatly weakened in complex situations. However, the neural network can extract high-order abstract features based on the original pixel data, which makes it possible to improve the effect of remote sensing image CD.

As mentioned earlier, the increase in remote sensing image information has been explosive growth, bringing the possibility for more technology applications in the field of CD [20], [21]. Deep learning technology can extract high features of remote sensing images with the support of a large number of data [22]–[25]. Thus, deeper difference representations can be learned by deep neural networks (DNNs), which can suppress the effect of noise and redundant information via layerwise nonlinear transformation [26]–[28] at the same time. However, DNN models rely heavily on large amounts of annotated training samples, which are typically generated manually with expert knowledge and tedious work. In the field of remote sensing images, very few open annotated datasets are available [29], [30]. Thus, remote sensing image processing can be regarded as a problem with few samples [31], [32], including the task of remote sensing image CD. Among them, the

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neural network models using a large proportion of unlabeled data to supplement the lack of supervised information are considered as semisupervised learning [33]–[35]. Meanwhile, emerging semisupervised approaches have been successfully applied to the image processing field using DNNs, such as object detection and semantic segmentation [36]–[38].

At present, some semisupervised methods based on deep learning have been used in the remote sensing image CD task, such as the generative adversarial network IR iteratively reweighted (GAN) network and the graph network. In the generative models, the discriminator is formed as a suitable feature extractor for the discriminative remote sensing image CD tasks, and the generator is used to simulate the information distribution of remote sensing image to generate false objects [39]–[41], while opportune generators and excellent semisupervised classifiers cannot be obtained simultaneously, impacting their applications in semisupervised learning tasks in practice [42], especially the remote sensing image CD task extremely lacking of data. Meanwhile, CD models based on graph networks use graph-based methods to construct structures connecting similar samples in the datasets [43], [44]. This leads to the sensitive penalty of their graph structures extracted from changed and unchanged regions in CD of the remote sensing image task. Those methods also need to analyze the complex time-consuming Laplacian matrix, which limits the remote sensing scope of these methods [45].

Being difficult to achieve a balance between discrimination and generation or sensitive to changeable graph structures, the existing semisupervised frameworks require a concise, fast but effective improvement urgently. Inspired by the ladder network [46], labeled data are expected to guide the similarity learning of unlabeled data directly to alleviate the unified training complex problem of the semisupervised remote sensing image CD task [47]. The direct joint learning of the labeled and unlabeled data reduces the training complexity of the network. However, the initial ladder network does not perform well in the complex remote sensing image CD task as even binary classification of remote sensing image CD can produce more changes than digital changes in the Mixed National Institute of Standards and Technology database (MNIST) datasets tested in the ladder network. The too-little-use of data leads to network learning not being sufficient and perfect. Therefore, the mode of circular incremental information supplement is expected with the detection performance gradually improving in a self-training way [48], [49].

Meanwhile, the existing semisupervised remote sensing image CD frameworks present fixed architectures, resulting the unimaginable burden of manual adjustment of the parameters once the dataset is changed. It is time-consuming and labor-consuming to redesign the network manually and leads to the semisupervised framework not being intelligent enough, so a way to search the optimal network structure automatically is attractive. Even if the datasets are switched, the network can adaptively achieve the best performance after the structural adjustment of the network. An effective way to obtain the optimal architecture of the neural network is structure searching based on the evolutionary algorithm, which can complete effective gene coding for the network in a fast heuristic search [50], [51]. As the previous task of CD for SAR images, Shi *et al.* [52] applied an efficient gene encoding to represent the network depth and the number of network nodes, which is capable of achieving effective and reasonable network architectures through mutation and other evolutionary operations. However, it is more inclined to choose a shallower network, which weakens the ability to utilize the high-order information in the middle layer extracted from the network.

In addition, the unsupervised analysis methods will be utilized to supplement the semisupervised classification through reusing the middle-layer feature information extracted by the network. The above semisupervised classification of CD is in a single perspective, which is deficient in stability. Thus, the unsupervised analysis is devised, which can obtain the predictive results, respectively, to assist the semisupervised network to judge the regional changes. It ensures the reliability of the pseudolabels to a greater extent by making judgments in the same region on many occasions.

In this article, a semisupervised adaptive ladder network (SSALN) for remote sensing image CD with label-incremental architecture searching is proposed, which can be summarized as the following three contributions.

1) We propose an SSALN for CD from two remote sensing images of any type, captured over the same scene at different times, which is capable to handle the issue of insufficient labeled data and avoid manually configuring the network. In SSALN, pseudolabels are updated iteratively. In each iteration, they are used together with limited real labels to automatically identify the most effective ladder network in terms of both architecture and parameters.
2) We devise a novel pseudolabel updating strategy that combines the strengths of semisupervised and unsupervised CD approaches, where unsupervised CD is applied to the output of each hidden layer in the main branch of the ladder network to generate multiple CD maps. These CD maps are then fused with the CD map generated by the semisupervisedly trained ladder network to produce the updated pseudolabels to be used in the next iteration.
3) We apply the proposed method to several pairs of homogenous images (e.g., both optical or both SAR images) and also several pairs of heterogenous images (e.g., optical and SAR images), leading to superior performance over existing CD techniques.

The rest of this article is organized as follows. Section II introduces the semisupervised CD problem and illustrates the motivation to address this problem. In Section III, the SSALN model is proposed and described in detail. The experimental results on several pairs of different remote sensing images and the related experiments on its rationality and practicability are discussed in Section IV. Finally, we make the conclusive remark and analysis in Section V.

II. Background and Motivation

A. Problem Statement

The rapid development of neural networks makes it possible to deal with the bursting growth of enormous remote sensing

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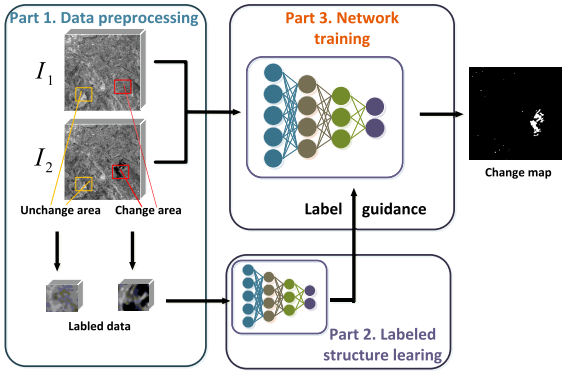


Fig. 1. Semisupervised CD in an unified way.

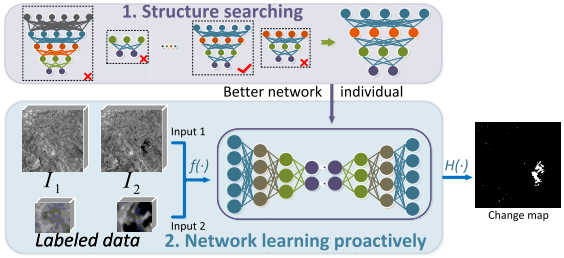


Fig. 2. Proactive learning with the dynamic structure in SSALN.

image data, which greatly improves the performance of CD. In the context of the extreme lack of labels, semisupervised methods have great potential in the field of remote sensing image CD.

The semisupervised remote sensing image CD problem can be formulated in Fig. 1. The processing flow of the remote sensing image CD task is divided into three parts.

1) *Data Processing:* The module of preprocessing remote sensing data.
2) *Labeled Structure Learning:* The module of training of labeled data in a model.
3) *Network Training:* The final module of synthesizing the untrained processed data, which then outputs the CM.

Two remote sensing images $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$ are processed, such as the operation of adjacent window pixel extraction in "Part 1." Then, labeled data sampled from changed and unchanged regions can be learned with additional structural analysis in "Part 2," which is then taken as the label guidance of the semisupervised training. Finally, the labeled structure learning in part two plays an auxiliary and guiding role in feature extraction, while it requires complex structure generation and additional structure analysis in "Part 3." After the guidance, all data are input into the network, and CD is completed through the final classification layer.

B. Motivation

The motivation is expected to be elaborated in three parts: additional complex structural analysis, fixed structure, and information supplement in lack of labels.

1) *Learn the Difference Information in Dual-Input Way Proactively:* As shown in Fig. 1, the existing semisupervised frameworks make the training of labeled data and unlabeled data relatively independent, difficult to train or to obtain good structure to represent remote sensing image information, which can be spread to unlabeled data. Thus, a concise but effective method is urgently needed.
   While the way of processing labeled and unlabeled data concurrently can effectively prevent additional structural analysis, it can promote unlabeled data to capture the clustering characteristics of labeled data quickly, which is shown in part two of Fig. 2 with dual-input way. Thus, higher order features are excepted to be gradually turned into more favorable for classification rather than reconstruction under the guidance of the limited labeled data. It makes network learning difference representation actively possible and the high-order features conducive to the remote sensing image CD task

$$\begin{aligned} CM &= \max(H_i[f(x, x_{\text{label}})]) \\ &= H_{\text{best}}[f(x, x_{\text{label}})] \end{aligned} \tag{1}$$

   where H_i is one of the network training individual and H_{best} is the selected better network training one. In the above processing method, unlabeled data and labeled data will be input into the network together, which is formed as $f(x, x_{\text{label}})$.
2) *Process in Dynamic Network Structure Toward Switched Data:* For the labels used for semisupervised network training being able to be provided by any pair of remote sensing images, it is naturally supposed that the semisupervised model is capable of turning into a general and universal form for switched detection objects. While toward the switches of detection objects, the difference information and processed data scale differ a lot.
   Therefore, a kind of self-adaptive structure needs to correspond with it, and one of the effective ways is to employ network architecture searching based on the evolutionary algorithm. In Fig. 2, the proposed method is divided into two parts to compare with the previously mentioned flow.
   a) *Structure Searching:* The evolutionary algorithm is applied to optimize the structures of networks. Through the continuous training of numerous networks in part two, the better network individual is obtained.
   b) *Network learning proactively:* Two remote sensing images $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$ and labeled data are simultaneously input into the ladder network. After the better network individual obtained, the final classification layer can output CD results.

Especially in multispectral remote sensing images, more spectrum bring abundant change information and different inputting data sizes after adjacent window extraction, leading to greater detection difficulty. Based on evolutionary structure searching, the trouble of manually adjusting the framework can be perfectly prevented.

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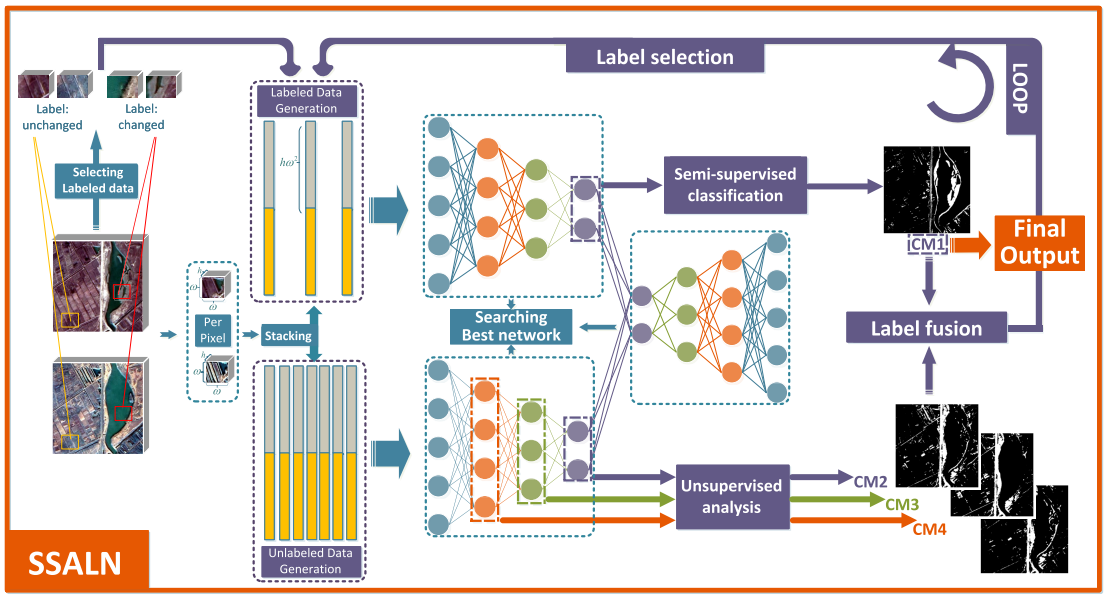


Fig. 3. Overview of the proposed SSALN, a semisupervised framework for the remote sensing image CD task.

3) *Promote the Circular Development of Available Labels in Network:* The change complexity of remote sensing images and the limitation of labels result in severe overfitting of the neural network to prediction.
To break through the limitation of restricted information, the outcome can be utilized more effectively by reusing feature information in the middle layer combined with the semisupervised output. A better network structure can be obtained in the proposed framework. The classification results will be obtained both from the unsupervised clustering and semisupervised network predicting in a self-circulation way.
Such two-predictive outcomes are helpful to supplement the classification information from different perspectives. Achieving a large number of reliable labels can obviously help the adaptive network to get better classification results. Besides, facing the problem of error accumulation of incorrect-classified labels in network retraining, it is a reliable solution to keep only the current training results.

In a word, a small number of change areas are excepted to be intuitively intercepted as samples with true labels and, finally, get the change results of remote sensing images through the processing of the adaptive model. It is a concise but effective semisupervised CD framework, and even if the detection object is switched, it can be well adaptive and compatible without manual modification of the network.

III. Methodology

In this section, the proposed SSALN in Fig. 3 will be described in detail:

1) *Subsection A:* The overall model will be summarized with circular and learning parts.
2) *Subsection B:* The dynamic network optimization based on evolutionary algorithms is discussed.

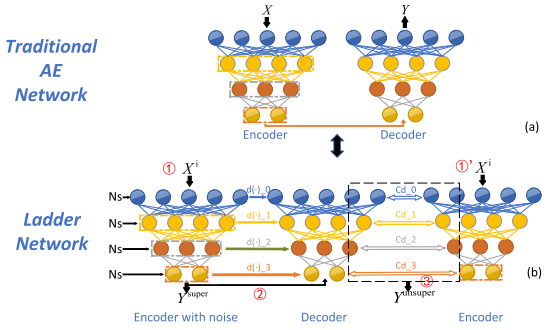


Fig. 4. Ladder network compared with traditional AE.

3) *Subsection C:* The label-selection strategy with detailed screening operation is talked in a combinatorial way.

A. Semisupervised Adaptive Ladder Network

1) Introduction of Autoencoder Network: It is worth mentioning that the proposed method is based on an autoencoder (AE) network.

Here, X is the input, and Y is the output in the AE net. Besides, in the ladder network, Ns is the adding noise, and X^i are all input data. Y^{super} is the classification result of labeled data, and Y^{unsuper} is the sum of unsupervised results on each layer.

AE net [53] is one of the most popular DNNs, which contains one encoder and one decoder, basing on the minimization of the error between the input X and reconstruction values Y in Fig. 4(a) and updating the weights and biases by the gradient descent method [54]. Nowadays, the AE network is widely applied in the field of remote sensing image processing. Geng *et al.* [55] designed a deep convolutional AE to extract representative features. Wang and Wang [56] proposed a stacked sparse AE with virtual adversarial regularization for PolSAR image classification.

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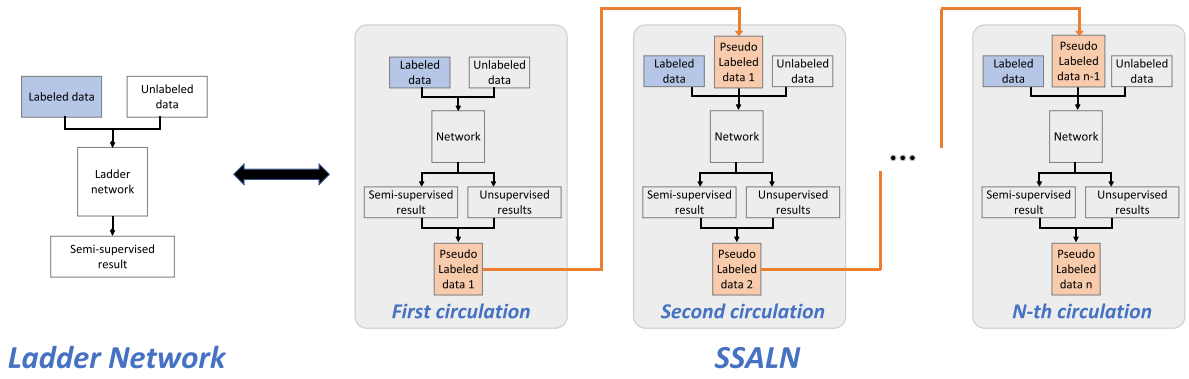


Fig. 5. Illustration of the overall circulation framework.

The ladder network is one variant of the AE network, which is shown in Fig. 4(b). Different from the traditional AE network, the structure of the ladder network is based on two encoders and one decoder. The steps involved in implementing the ladder network are typically given as follows.

1) Take an encoder model with adding noise on each layer as supervised learning between labeled data, which is connected to a classification layer.
2) Reconstruct in a decoder with denoising work $d(\cdot)_i$, which inverts the mappings on each layer of the encoder.
3) Restore in an encoder without noise with one-to-one correspondence Cd_i to the decoder as unsupervised learning between all data.
4) Minimize the sum of all the cost function terms with both supervised and unsupervised learning.

2) Introduction of SSALN: Based on the ladder network, which is the "Network" part in every circulation in Fig. 5, the proposed framework makes use of the unsupervised restore part in the net.

1) It is then divided into the supervised part and the unsupervised part. In the supervised part, the labeled data will be classified by connecting the classification layer after the feature extraction of the encoder with noise.
2) The results of the supervised part will be fused with the results of the unsupervised part, in order to generate the pseudolabeled dataset for circulation training.

Based on the hypothesis of low density, the proposed framework aims to get similar classified output from input with similar change attributes, with noise added to unlabeled data, and minimizing reconstruction error through the reconstruction Cd_i of labeled data on each layer in the ladder network [46]. Noise interference $d(\cdot)_i$ on each layer makes the model insensitive to the disturbance of the training data, which means that the decision boundary is encouraged to be in the low-density region.

The initial ladder network is tested on MNIST and Canadian Institute for Advanced Research (CIFAR)-10 datasets, performing not good in remote sensing image CD. Because the structure and shape of the number are basically the same, neural network can extract its important features, but the regional change of remote sensing images is relatively complex, and the initial classification accuracy is often unsatisfactory. Due to the complexity of change difference of remote sensing images, too few limited numbers of labels in the semisupervised network always results in overfitting of the outcome.

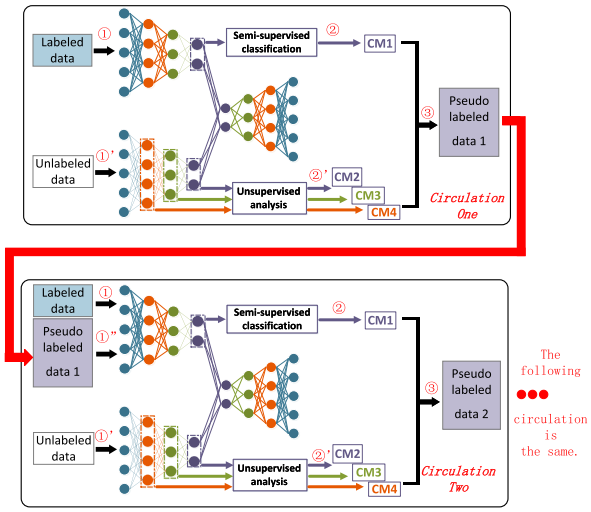


Fig. 6. Visual display of sample selection.

Thus, circular information reuse on outcomes of change map produced in semisupervised network and unsupervised method means to fully utilize the existing classification structure features. A combinatorial label-selection strategy ensures the reliability of predictive classification. With the incremental supplement of potential information, the performance of distinguishing differences has been significantly improved.

3) Semisupervised Annotation Case: In SSALN, pseudolabels are updated iteratively. As shown in Fig. 6, after the semisupervised annotation, the labels have been updated.

1) Labeled data and unlabeled data are inputted into the network.
2) Semisupervised and unsupervised results can be obtained.
3) After the fusion of two kinds of results such as CM1(Change Map 1), CM2, CM3, and CM4, the pseudolabeled data are selected.

In circulation, the pseudolabeled data will be added into the part of labeled data, which is regarded as the third kind of

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input data in step "Pseudolabeled data 1" in (b) of Fig. 1. Such two-predictive outcomes in semisupervised and unsupervised ways are helpful to supplement the classification information in different perspectives.

4) Circular Sample Selection: There is an issue that the number of labels manually labeled is too small to fully summarize the characteristics of the changed and unchanged regions of the remote sensing image. Therefore, why not make more use of our classified labels to expand our prior knowledge?

Thus, the amount of information is expected to be enriched through the procession of two different perspectives, where the first prediction results of the network will be fused with the several classification results obtained by the unsupervised method. Then, the principle of "the minority is subordinate to the majority" is adopted to screen the relatively reliable pseudolabels in order to increase the confidence degree of the labels. After that, the network is retrained by the iterative operation.

Finally, with the increase in available labeled data, the repredictive result is undoubtedly more reliable. Of course, even if the obtained labels are more reliable, there is still the possibility of error. In order to prevent the continuous interference of the wrong-divided labels on the network, only the last network prediction result will be reused. Surprisingly, most of the remote sensing image labels can be obtained by the first sample selection, which is reliable proved by experiments.

As shown in Fig. 6, the labels selected under multilayer monitoring are believed to be reliable. In this way, more pseudolabels, in turn, guide the training of the network again. Rich prior knowledge can undoubtedly improve the classification performance of the network in a self-training way. In addition, if accurate results are not necessary, the circular part of the network can be abandoned to complete the fast CD operation.

5) Learning of Framework: In the proposed framework, the circular structure plays an important role in the classification task. Algorithm 1 shows the learning process of SSALN.

In the proposed iterative framework, the losses L over all time steps $t \in \{t|t = 1, 2, \ldots, T\}$ can be accumulated and formulated as

$$\min_{l,n,\{\lambda_i\}} L(x; x_{\text{label}}) = \sum_{t=1}^{T} L^t(x; x_{\text{label}}). \tag{2}$$

At stage t, the loss function of SSALN can be formulated as follows (it is worth noting that, at the final stage, noise n will not be added in network training):

$$\begin{aligned} L^t(x; x_{\text{label}}) &= C_c + C_d \\ &= C_c + \sum_{i=0}^{l-1} \lambda_i \cdot C_d^i \\ &= -\frac{1}{n} \sum_{i=1}^{n} \log P(g(x_{\text{label}})|y_{\text{label}}) \\ &\quad + \frac{1}{n} \sum_{i=0}^{l-1} \frac{\lambda_i}{m_i} \sum_{j=1}^{n} \left\| f(x)_i^j - (x+n)_i^j \right\|^2 \end{aligned} \tag{3}$$

Algorithm 1 Learning Process of SSALN

Input: X:Remote image datasets; X_{all}:all data; $X_{labeled}$:data with true labels; $iter_{max}$:the number of iterations.
Output: CM: change map.
Step 1) Data X adjacent-window processing.
Step 2) Preparation of X_{all} and $X_{labeled}$.
Step 3) while $i < iter_{max}$ **do**
  if $i > 1$ **then**
    $X_{labeled} = X_{labeled} + X_{pseudo}$
  i. Ladder network training;
  ii. Outputting semisupervised CM: CM_{semi};
  iii. Extracting hidden-layers information;
  iv. Outputting unsupervised CMs: CM_i;
  v. CMs fusion and selecting pseudo labels:
    $X_{pseudo} = CM_{semi} \cap CM_i$;
  vi. $i = i + 1$;
  end while
Step 4) return The final semisupervised CM.

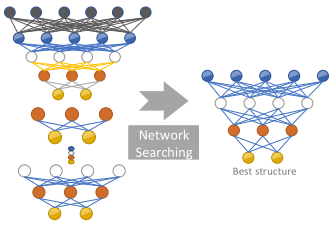


Fig. 7. Illustration of network searching.

where C_c is the supervised loss, C_d is the unsupervised loss in each layer, n is the number of training samples, and $P(\cdot)$ is the average negative log probability of the noisy output $g(x_{\text{label}})$ given the prior labels y_{label}. l is the number of network layers, λ_i is the unsupervised penalty factor for each layer, and m_i is the number of nodes per layer. $f(x)_i^j$ is the unsupervised output, and x_i^j is the processed pixel pair on each sample j of each layer i.

Based on the above, a semisupervised circulative remote sensing image CD framework is proposed, which only uses a small number of manually identified labels to obtain the final change map. In addition, we try to test the classification performance with the minimum number of labels, which is gotten in advance, and its practicality is impressive.

B. Dynamic Framework With Self-Variable Structure

The introduction of structure searching is more to solve the problems of large-scale, long-time, and low efficiency of manual parameter adjustment, and the pseudocodes describe the flow of adaptive structure training in circular networks. As shown in Fig. 7, a better structure can be obtained after self-adaptive searching. The whole network structure searching is based on the evolutionary algorithm, which employs the special combinatorial offspring selection method toward intermediate information utilization for multiple hidden layers.

1) Dynamic Network Structure: In the process of evolutionary selection, only the number of network layers and nodes

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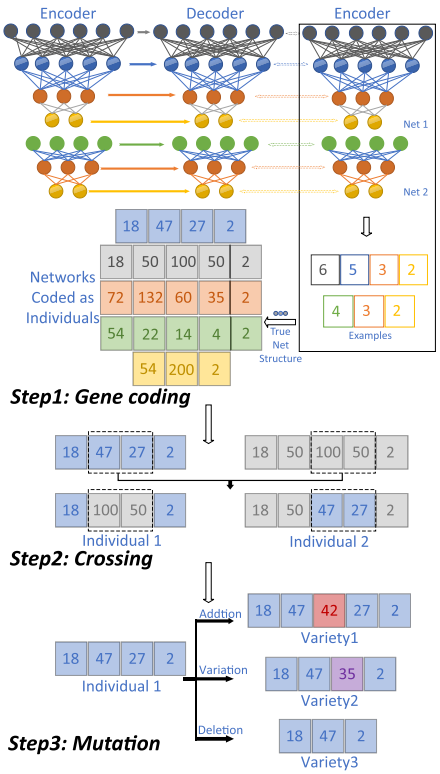


Fig. 8. Population evolution of the network.

will be searched, so as to reduce the whole evolutionary searching space.

The number of network layers and nodes for the coding layer of the AE network are encoded in the network. The decoding layer corresponds to the coding layer one by one, so no additional coding operation is required, which is shown in step 1 of Fig. 8, taking two nets as examples. For net 2, it contains four, three, and two nodes in its first, second, and third layers, so it is coded as [4, 3, 2]. In Algorithm 2, the initial population of SSALN individuals is disposed of different numbers of layers and different numbers of nodes in each layer; then, the corresponding evolutionary operation, such as crossing and mutation in step 2 and step 3 of Fig. 8, is carried out; finally, the offspring individuals are selected with the highest classification accuracy. In addition, each circulation of the network will search the neural architecture again because more information on predictive labels will also bring about inestimable changes.

2) Fitness Evaluation Combined With Average Unsupervised Loss: In the structure searching, a large number of network individuals are generated, where criteria should be designed to evaluate the advantages and disadvantages of network individuals. Both the average level α, the total loss L^t, and the loss of unsupervised layers L_u^t will be considered. First, α is used to evaluate basic merit index of two individuals, which ensures that each selected individual is above the average level

$$\alpha = \left(L^t(x; x_{\text{label}})_{\max} - L^t(x; x_{\text{label}})_{\min}\right)/N \tag{4}$$

where L^t is the loss of step t and N represents the number of network individuals.

Algorithm 2 Dynamic Compatible Structure Adjustment

Input: X:Remote image datasets; n:popsize; $iter_{max}$:the number of iterations.
Output: The better architecture.
Step 1) Population initialization.
Step 2) Calculate fitness L^t of each individual.
Step 3) while $i < iter_{max}$ **do**
1: **i.** Generation of offsprings;
2: **ii.** Comparative fitness;
3: **if** no better elimination **then**
4: **break**
5: **iii.** Evolutionary selection;
6: **iv.** $i = i + 1$;
7: **end while**
Step 4) return Better architecture.

Besides, the high-order features will be applied for unsupervised analysis to supplement semisupervised analysis. Thus, unsupervised attributes L_u^t are additionally considered in the selection to guarantee the amount of labeled information in the unsupervised part of the model

$$L_u^t = \sum_{i=0}^{l-1} C_d^i \Big/ l \tag{5}$$

$$\text{better individual} = \begin{cases} A, & (L^t)_B - (L^t)_A \geq \alpha \\ & \text{or} \\ & (L^t)_A - (L^t)_B \geq \alpha \,\&\, (L_u^t)_B \geq (L_u^t)_A \\ B, & \text{otherwise.} \end{cases} \tag{6}$$

In (6), the overall loss L^t of individual B is greater than individual A by default. Otherwise, the positions of individuals A and B will be swapped. When $((L^t)_B - (L^t)_A)$ is less than the average level α, the individual A with a lower unsupervised loss $(L_u^t)_A$ still tends to be chosen, which means that the better reconstruction classification is preferred. Algorithm 3 reflects the whole process of evolutionary selection.

3) Prevention of Population Degradation: In order to prevent meaningless network evolution, the principle of "stop without better offspring" is adopted. This means that, if the classification performance of the offspring in the next three generations of the network is not as good as that of the worst network individual in the parent generation, the circulation will be directly stopped, and the optimal network individual will be retained at this time.

C. Combinatorial Label-Selection Strategy

1) Unsupervised Clustering Operation: In addition to the direct prediction results of the initial network, a series of high-order features extracted will be reused. It mentions that a simple unsupervised AE network is more inclined to extract features conducive to reconstruction and restoration, while the dual-input label-incremental network can extract high-order features are more inclined to classification under the guidance of labeled data, whose attributes are changed.

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Algorithm 3 Combinatorial Strategy of Evolution Under Unsupervised Control

Input: p:Crossover rate; q:variation rate; pop:Network population.
Output: Optimal individual.
Step 1) Crossing and mutation to get offspring off_{pop}.
Step 2) Calculate the fitness value L^t of each offspring, including total loss a and average unsupervised loss b.
Step 3) Calculation $\alpha = (L^t{}_{max} - L^t{}_{min}) \,/\, n$.
Step 4) **if** $a_1 - a_2 > \alpha$ **then**
1: **return** a_2
2: **else**
3: **if** $b_1 > b_2$ **then**
4: **return** a_2
5: **else**
6: **return** a_1
7: **end**
8: **end**
Step 5) **return** Optimal individual.

Next, the unsupervised clustering method is applied to the high-order features extracted from each layer, which suppresses the influence of redundant information and noise, and more suitable for classification. Finally, for the feature extraction results of each layer, the corresponding change map can be obtained so that the classification results can be achieved from the unsupervised perspective.

2) Selection of new Predictive Labels: For each layer of unsupervised prediction results nl and semisupervised network results nl_{semi}, the same percentage of points will be given. Thus, the total score NL_j can be formed as follows:

$$(NL)_j = \sum_{i=0}^{l-1} nl_i + nl_{\text{semi}} \tag{7}$$

where l is the number of network layers, and the scores for the changed and unchanged areas are given as follows:

$$nl_i, nl_{\text{semi}} = \begin{cases} 0, & \text{predictive unchanged} \\ 1, & \text{predictive changed.} \end{cases} \tag{8}$$

The following rules are employed to filter for reliable changed and unchanged pseudolabels:

$$(NL)_j = \begin{cases} l, & \text{changed sample} \\ 0, & \text{unchanged sample} \\ \text{otherwise}, & \text{uncertain sample.} \end{cases} \tag{9}$$

In this way, the sample selection from the perspective of semisupervised network prediction and unsupervised clustering classification could be obtained, and then, the network will be trained again to achieve the purpose of increasing the accuracy. However, even if we have multilayer control, there may still be incorrect-divided samples and then flow into the whole network. Thus, the sample selection will be only used after the last screening.

TABLE I
INTRODUCTION OF TEST DATA

Remote sensing image types	Dataset	Pixel size
SAR	Bern	301 × 301
	Ottawa	290 × 350
	Yellow River	450 × 280
Optical	Mexico City	512 × 512
	Sardinia	412 × 300
Heterogeneous	Wuhan	503 × 495
	Sardinia	412 × 300
	Shuguang	921 × 593
VHR	Guangzhou	877 × 738
	Montpellia	451 × 426
	Shanghai	873 × 782
Multispectral	Hongqi	539 × 543
	Minfeng	651 × 461
	Yandu	322 × 266

IV. EXPERIMENTAL STUDY

In this section, the experiments will be explained systematically in detail:

1) *Subsections A and B:* The experimental setup is introduced.
2) *Subsection C:* To validate the superiority and universality, a variety of remote sensing image datasets are tested on the proposed framework compared with other methods.
3) *Subsection D:* The anti-interference ability of the model is tested on the uncalibrated images additionally.
4) *Subsections E and F:* For the proof of the validity of the network circulation and structure searching parts in the framework, a comparative experiment is done.
5) *Subsection G:* To prove the practicability of the model of CD, the experiment on the minimum number of labels is carried out.

A. Data Introduction

The datasets include five types of remote sensing images in Table I; each pair of them has been radiometrically corrected and coregistered to make them as more comparable as possible. The test indicators will be shown in the following tables of each remote sensing image dataset type.

In our experimental tests, the percentage of correct classification (PCC) [8] and the kappa coefficient [57] are used to quantify the performance of remote sensing image CD.

B. Experiment Setting

The whole framework is based on one-time network training and circulative part. Some detailed parameters will be described as follows.

1) One-Time Network Training: The remote sensing images are extracted in a local neighborhood size of $w \times w \times h$ as one sample at each position, and h represents the number of bands in the image. 100 pixels are selected as semisupervised training labels. The noise factor for AE part is set to be 0.3, and the reconstruction layer weights are set to be [1000.0, 0.10, 0.010, 0.0010, 0.00010].

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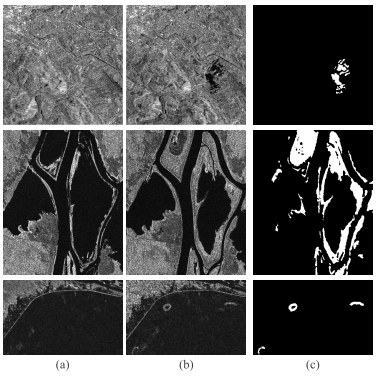


Fig. 9. (a) SAR image I_1. (b) I_2. (c) Reference map.

2) Circulative Training: In order to prevent overfitting and learn the variable difference information proactively in switched detection objects in remote sensing images, the algebra of structure searching is lower than that of complete network training. While facing the design of network structure itself, the too large and complex neural network is not advocated. The ideal network structure should have few nodes in each layer and tend to be deeper for the multilayer unsupervised feature extraction.

C. Performance of SSALN Compared With Other Methods

In experiments, all performance comparisons will be divided into three categories: unsupervised, semisupervised, and supervised to help understand the detection ability relative to labeled and unlabeled methods. Besides, for the supervised way, we added the comparison of ResNet [58] with fully connected structure applied on remote sensing images [59] as one supervised algorithm, which used 100, 60%, and 100% labeled data, respectively. At the same time, we take SSALN with the same 60% labeled data as the semisupervised method under the unified standard to compare the performance. Besides, to meet the fixed limited labels, we only elaborate SSALN(100) in the experimental analysis part.

1) SAR Image: For SAR remote sensing images, Fig. 15 shows Bern, Ottawa, and Yellow River datasets from top to bottom compared with reformulated fuzzy local information C-means (RFLICM) [14], multiscale superpixel segmentation DNN (MSSDNN) [60], evolving fully connected neural network (EVONN) [52], and Semi-GAN [40] methods. Index results are shown in Table II. Fig. 9 presents the related SAR image dataset.

Subjectively, the detection performance of each method is satisfactory, which is due to the less internal change structure of the Bern dataset and the simplicity of the SAR image. For the Ottawa dataset, all the methods have detected the main change areas of the Ottawa dataset, but the main difficulty of the dataset is that there are a small number of unchanged points in the middle of a large number of change areas, especially in the upper left corner of a large area of change region in Fig. 15(f), while the proposed method ensures extremely fine classification performance. The Yellow River dataset is influenced by noise much greatly, making it hard to detect the changes occurring on it. However, in the comparison of PCC and kappa coefficients, the proposed method has achieved the best results, and at the same time, it is very successful to restore the details of the lower left corner, performing better than the methods of DNN and EVONN in Column 3 of Fig. 15.

TABLE II
EVALUATION INDEXES OF THE SAR IMAGE

Dataset	Methods		PCC	Kappa
Bern	Unsupervised	RFLICM	0.9964	0.8377
		MSSDNN	0.9960	0.8545
		EVONN	0.9965	0.8589
	Semi-supervised	Semi-GAN(100)	0.9947	0.8073
		SSALN(100)	0.9964	0.8634
		SSALN(60%)	**0.9970**	**0.8838**
	Supervised	Resnet(100)	0.1878	0.0841
		Resnet(60%)	0.9871	0.8173
		Resnet(100%)	0.9960	0.8434
Ottawa	Unsupervised	RFLICM	0.9733	0.8935
		MSSDNN	0.9719	0.8981
		EVONN	0.9845	**0.9845**
	Semi-supervised	Semi-GAN(100)	0.9738	0.8994
		SSALN(100)	0.9851	0.9444
		SSALN(60%)	**0.9889**	0.9588
	Supervised	Resnet(100)	0.1096	0.0528
		Resnet(60%)	0.9576	0.8771
		Resnet(100%)	0.9736	0.9029
Yellow River	Unsupervised	RFLICM	0.9878	0.5998
		MSSDNN	0.9669	0.3567
		EVONN	0.9963	0.7999
	Semi-supervised	Semi-GAN(100)	0.9928	0.5933
		SSALN(100)	0.9975	0.8778
		SSALN(60%)	**0.9977**	**0.8901**
	Supervised	Resnet(100)	0.2578	0.0190
		Resnet(60%)	0.8593	0.1732
		Resnet(100%)	0.9726	0.4142

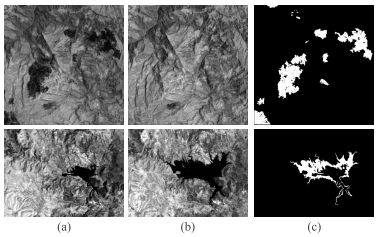


Fig. 10. (a) Optical image I_1. (b) I_2. (c) Reference map.

Besides, SSALN using 100 labeled training data already achieves a very good result, and when it is provided more training samples (i.e., 60% of the total available labeled data), its performance gets further improved. Notably, our proposed method can even outperform the supervised ResNet using 100% of the available labeled data for training. SSALN is the combination of supervised and unsupervised ways, which is capable to make full use of enormous unlabeled information than the supervised method (i.e., Resnet). Moreover, supervised net under the 100% labels is also prone to emerging problems, such as overfitting, which leads to worse results.

2) Optical Image: For optical remote sensing images, Fig. 16 shows Mexico City and Sardinia datasets from top to bottom compared with symmetric convolutional coupling network (SCCN) [61], EVONN [52], and Semi-GAN [40] methods. Index results are shown in Table III. Fig. 10 presents the related optical image dataset.

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TABLE III
EVALUATION INDEXES OF THE OPTICAL IMAGE

Dataset	Methods		PCC	Kappa
Mexico City	Unsupervised	SCNN	0.9826	**0.9011**
		EVONN	0.9910	0.8247
	Semi-supervised	Semi-GAN(100)	0.9731	0.4763
		SSALN(100)	0.9933	0.8785
		SSALN(60%)	**0.9934**	0.8833
	Supervised	Resnet(100)	0.1584	0.1453
		Resnet(60%)	0.9382	0.7354
		Resnet(100%)	0.9716	0.8571
Sardinia	Unsupervised	SCNN	0.9618	0.6796
		EVONN	0.9546	0.7007
	Semi-supervised	Semi-GAN(100)	0.9601	0.7219
		SSALN(100)	0.9836	0.8702
		SSALN(60%)	**0.9905**	**0.9205**
	Supervised	Resnet(100)	0.1852	0.2399
		Resnet(60%)	0.9359	0.6801
		Resnet(100%)	0.9816	0.8602

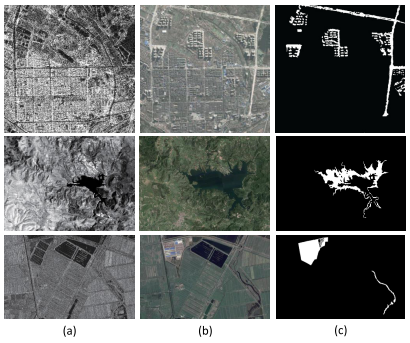


Fig. 11. (a) Heterogeneous image I_1. (b) I_2. (c) Reference map.

As can be seen, the proposed method has the lowest false detection rate in Table III. It deals well with the boundary between the changed and unchanged regions with the overfitting of the unchanged area being more slight in the Mexico City dataset. For the Sardinia dataset, it can be obviously seen that the proposed method is superior to other detection methods with a clearer outline in Fig. 16(d), which is also proven that the detection coefficients of PCC and Kappa are ranked first in the Sardinia dataset of Table III.

3) Heterogeneous Image: For heterogeneous remote sensing images, Fig. 17 shows Wuhan, Yellow Sardinia, and Shuguang datasets from top to bottom compared with adaptive local structure consistency (ALSC)-O [62], patch similarity graph matrix (PSGM) [63], and Semi-GAN [40] methods. Index results are shown in Table IV. Fig. 11 presents the related heterogeneous image dataset.

For the Wuhan dataset, the proposed method can suppress the noise interference to a certain extent, it also leads to the weakening of the detection performance in the main change areas, and that is why it gets a lower score in the coefficient of Kappa in Table IV. Around the changed area of Sardinia, all methods inevitably appear the problem of noise. The proposed method can suppress the noise in a large unchanged area, but this is achieved at the expense of inadequate detection of changed regions in Fig. 17(e). Overfitting and underfitting appear simultaneously in the Shuguang dataset. Visually, there is no significant difference between the methods. On the Kappa coefficients, the proposed method is slightly lower than the PSGM method in the Shuguang dataset of Table IV, for there being more discontinuous black false detection area in Fig. 17(b).

TABLE IV
EVALUATION INDEXES OF THE HETEROGENEOUS IMAGE

Dataset	Methods		PCC	Kappa
Wuhan	Unsupervised	ALSC-O	0.9540	0.5950
		PSGM	0.9532	0.6858
	Semi-supervised	Semi-GAN(100)	0.8973	0.3957
		SSALN(100)	0.9579	0.6556
		SSALN(60%)	**0.9725**	**0.7113**
	Supervised	Resnet(100)	0.1985	0.0921
		Resnet(60%)	0.8257	0.4897
		Resnet(100%)	0.9124	0.5393
Sardinia	Unsupervised	ALSC-O	0.9620	0.6983
		PSGM	0.9610	0.6821
	Semi-supervised	Semi-GAN(100)	0.8867	0.3908
		SSALN(100)	0.9670	0.7115
		SSALN(60%)	**0.9842**	**0.8571**
	Supervised	Resnet(100)	0.2091	0.1204
		Resnet(60%)	0.9197	0.4981
		Resnet(100%)	0.9423	0.6307
Shuguang	Unsupervised	ALSC-O	0.9580	0.6410
		PSGM	0.9766	0.7438
	Semi-supervised	Semi-GAN(100)	0.9257	0.4629
		SSALN(100)	0.9777	0.7295
		SSALN(60%)	**0.9795**	**0.7522**
	Supervised	Resnet(100)	0.2402	0.2116
		Resnet(60%)	0.9155	0.5901
		Resnet(100%)	0.9498	0.6142

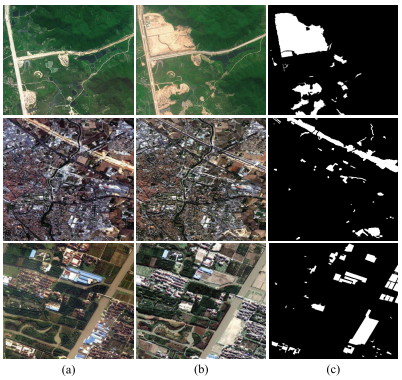


Fig. 12. (a) VHR image I_1. (b) I_2. (c) Reference map.

4) VHR Image: For VHR remote sensing images, Fig. 18 shows Guangzhou, Montpellia, and Shanghai datasets from top to bottom compared with deep CVA (DCVA) [64], deep belief network (DBN) [65], multiscale difference representation learning (MSDRL) [66], and Semi-GAN [40] methods. Index results are shown in Table V. Fig. 12 presents the related VHR image dataset.

For the middle three small parts of the change regions, only the proposed method achieves the best detail retention ability, which is why it obtains the highest PCC and kappa coefficients in the Guangzhou dataset of Table V. Compared with the Guangzhou dataset, the Montpellia dataset has too many changed scattered regions, which leads to the phenomenon of overfitting in Column 2 of Fig. 18. In the Shanghai dataset, the highest PCC and kappa coefficients prove that the

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TABLE V
EVALUATION INDEXES OF THE VHR IMAGE

Dataset	Methods		PCC	Kappa
Guangzhou	Unsupervised	DCVA	0.9606	0.8343
		DBN	0.9666	0.8572
		MSDRL	0.9752	0.9007
	Semi-supervised	Semi-GAN(100)	0.9247	0.6654
		SSALN(100)	0.9781	0.9118
		SSALN(60%)	**0.9796**	**0.9189**
	Supervised	Resnet(100)	0.3914	0.2985
		Resnet(60%)	0.9385	0.8571
		Resnet(100%)	0.9701	0.8873
Montpellia	Unsupervised	DCVA	0.9425	0.6101
		DBN	0.9514	0.5557
		MDSRL	0.9694	0.7339
	Semi-supervised	Semi-GAN(100)	0.9118	0.2977
		SSALN(100)	0.9591	0.6404
		SSALN(60%)	**0.9709**	**0.7424**
	Supervised	Resnet(100)	0.2319	0.2310
		Resnet(60%)	0.8692	0.4932
		Resnet(100%)	0.9248	0.5741
Shanghai	Unsupervised	DCVA	0.9233	0.5729
		DBN	0.9339	0.5145
		MSDRL	0.9524	0.6768
	Semi-supervised	Semi-GAN(100)	0.8889	0.5010
		SSALN(100)	0.9553	0.7117
		SSALN(60%)	**0.9714**	**0.8144**
	Supervised	Resnet(100)	0.3981	0.1953
		Resnet(60%)	0.8962	0.6294
		Resnet(100%)	0.9386	0.6977

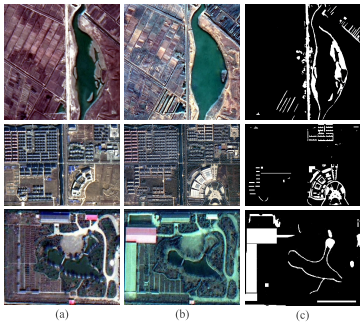


Fig. 13. (a) Multispectral image I_1. (b) I_2. (c) Reference map.

TABLE VI
EVALUATION INDEXES OF THE MULTISPECTRA IMAGE

Dataset	Methods		PCC	Kappa
Hongqi	Unsupervised	IR-MAD	0.9491	0.6856
		DNN	0.9532	0.7231
		GAND	0.9476	0.7252
		GDCN	0.9697	0.8107
	Semi-supervised	Semi-GAN(100)	0.9077	0.4846
		SSALN(100)	0.9713	0.8297
		SSALN(60%)	**0.9751**	**0.8491**
	Supervised	Resnet(100)	0.3091	0.2712
		Resnet(60%)	0.8593	0.6767
		Resnet(100%)	0.9439	0.7292
Minfeng	Unsupervised	IR-MAD	0.9376	0.5221
		DNN	0.8810	0.4531
		GAND	0.9533	0.6151
		GDCN	0.9526	0.6421
	Semi-supervised	Semi-GAN(100)	0.9269	0.4529
		SSALN(100)	0.9563	0.6668
		SSALN(60%)	**0.9679**	**0.6714**
	Supervised	Resnet(100)	0.1842	0.2011
		Resnet(60%)	0.8875	0.5491
		Resnet(100%)	0.9243	0.6063
Yandu	Unsupervised	IR-MAD	0.9141	0.6817
		DNN	0.9446	0.7693
		GAND	0.9449	0.7538
		GDCN	0.9058	0.7909
	Semi-supervised	Semi-GAN(100)	0.9058	0.6308
		SSALN(100)	0.9523	0.7978
		SSALN(60%)	**0.9540**	**0.8029**
	Supervised	Resnet(100)	0.3099	0.2549
		Resnet(60%)	0.8591	0.6771
		Resnet(100%)	0.9279	0.7255

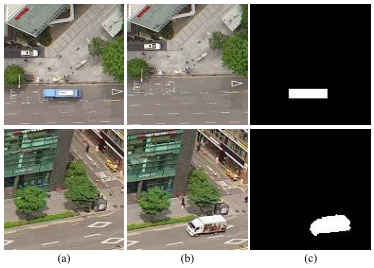


Fig. 14. (a) UAV image I_1. (b) I_2. (c) Reference map.

proposed method is superior to other methods, and the results are consistent subjectively in Table V.

5) Multispectra Image: For multispectral remote sensing images, Fig. 19 shows Hongqi, Minfeng, and Yandu datasets from top to bottom compared with IR-multivariate alteration detection (MAD) [67], DNN [68], GAND [69], generative discriminatory classified network (GDCN) [70], and Semi-GAN [40] methods. Index results are shown in Table VI. Fig. 13 presents the related multispectra image dataset.

From the detection results, it can be seen that the proposed method can keep the best detail in the whole remote sensing image and suppress the noise in the middle area, achieving the best PCC and kappa coefficients in Table VI. Compared with the Hongqi dataset, the CD of the Minfeng dataset is more difficult. As can be seen, the proposed method achieves better performance, outperforming other CD methods with the highest PCC and kappa coefficients in the Minfeng dataset of Table VI. As shown in Fig. 19(f), a good detection result is obtained in the unchanged area in the middle of the changed area. For the Yandu dataset, the too-strong-fitting effect weakens the detection effect of the middle road, which looks like a clothes rack in Fig. 19(e). As can be seen, the proposed method achieves the best CD results with the highest PCC and kappa coefficients.

D. Extra Comparison of Uncalibrated Objects: UAV as an Example

Previous remote sensing images are all matched and aligned, so there is no extra pixel-pair misalignment. However, for UAV images obtained by the camera mounted on the unmanned aerial vehicle, it comes from different view positions and angles. Therefore, whether the proposed semisupervised framework can still complete the detection task under the guidance of few labels of the unregistered images is worth discussing.

Fig. 14 presents the related UAV image dataset, and Fig. 20(a)–(g) shows the CD results of the UAV dataset compared with classification-based [71], Superpixel-based [72],

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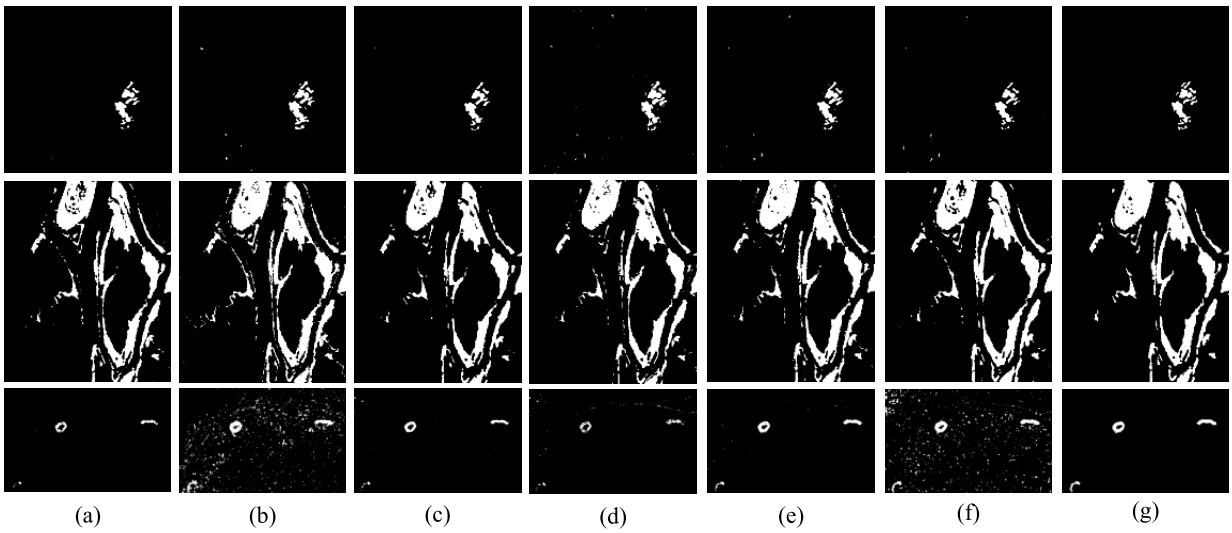


Fig. 15. SAR images datasets and CD results. (a) DNN. (b) MSSDNN. (c) EVONN. (d) Semi-GAN. (e) SSALN(100). (f) Resnet(100%). (g) Reference map.

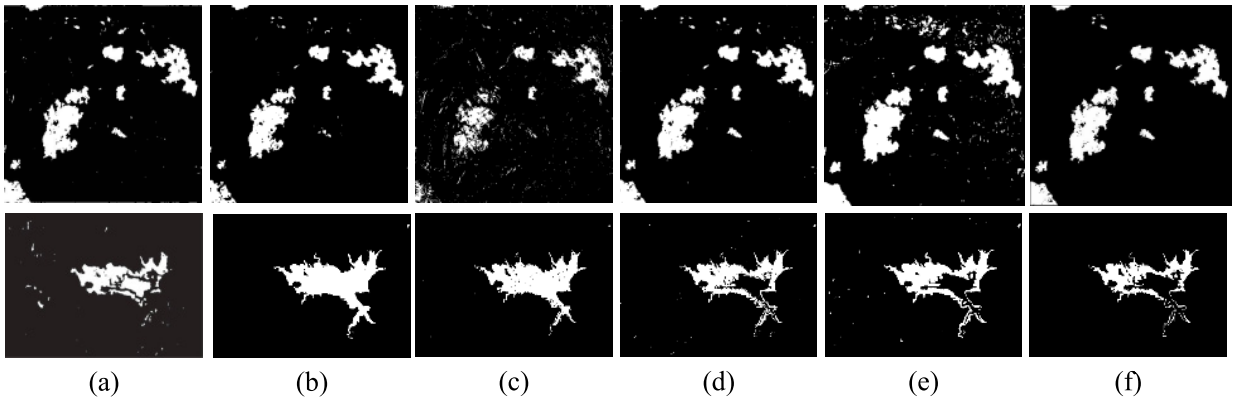


Fig. 16. Optical images datasets and CD results. (a) SCCN. (b) EVONN. (c) Semi-GAN. (d) SSALN(100). (e) Resnet(100%). (f) Reference map.

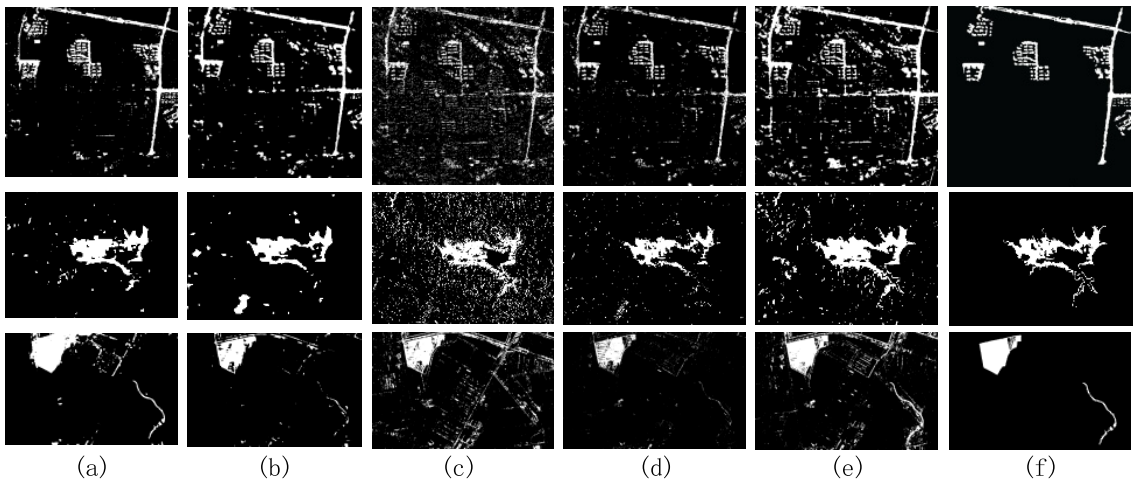


Fig. 17. Heterogeneous images datasets and CD results. (a) ALSC-O. (b) PSGM. (c) Semi-GAN. (d) SSALN(100). (e) Resnet(100%). (f) Reference map.

and bipartite differential neural network (BDNN) [73] methods. It can be seen from the results that the performance of the traditional methods in Fig. 20(b), (d), and (f) is not good, remaining many kinds of interference for it is limited to a consistent coordinate system. While the BDNN method adopts the way of content comparison, which can extract the holistic features from the unchanged regions, it has achieved good results.

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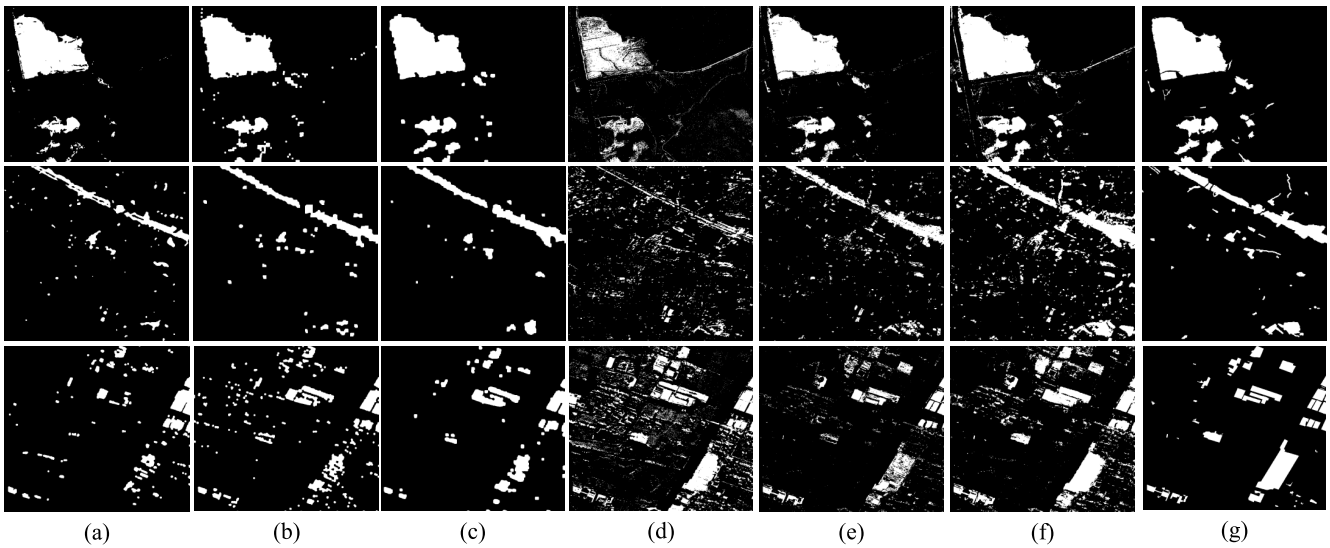


Fig. 18. VHR images datasets and CD results. (a) DCVA. (b) DBN. (c) MSDRL. (d) Semi-GAN. (e) SSALN(100). (f) Resnet(100%). (g) Reference map.

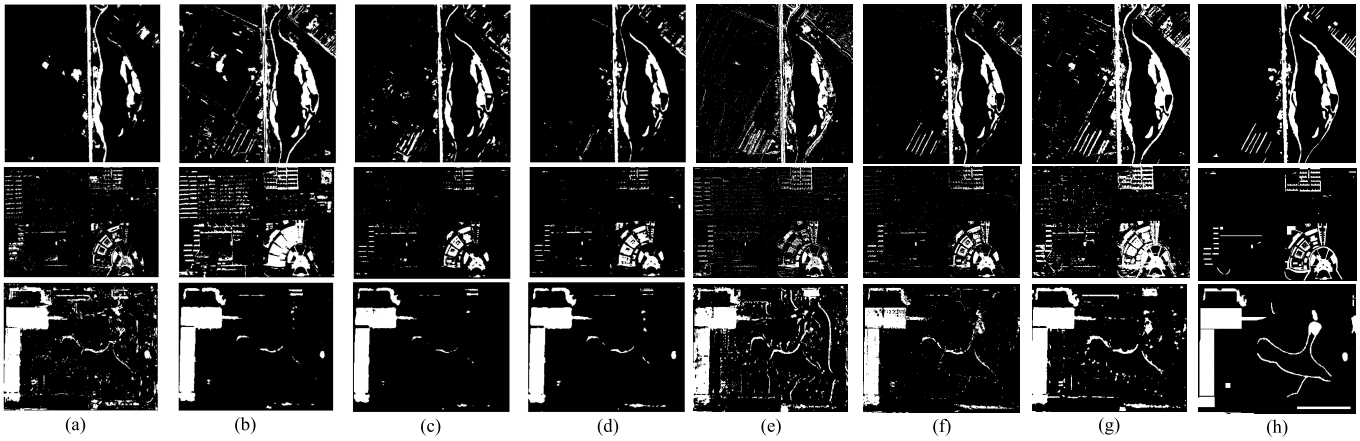


Fig. 19. Multispectral images datasets and CD results. (a) IR-MAD. (b) DNN. (c) GAND, (d) GDCN. (e) Semi-GAN. (f) SSALN(100). (g) Resnet(100%). (h) Reference map.

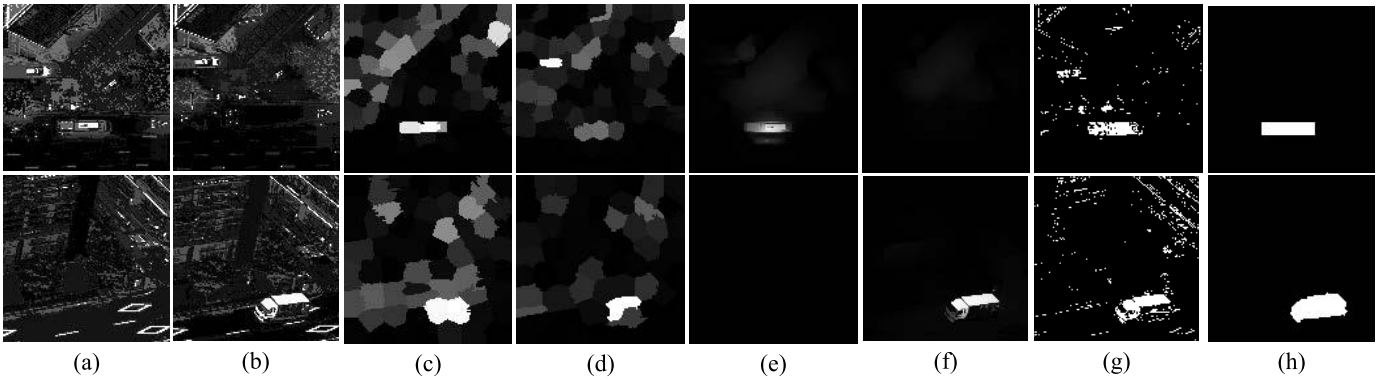


Fig. 20. Difference images generated on the UAV dataset by (a) and (b) classification-based, (c) and (d) Superpixel-based, and (e) and (f) BDNN and CD result achieved by (g) SSALN and (h) reference map.

However, it is worth mentioning that, compared with the special-designed BDNN for UAV image, the proposed method can mostly suppress the surrounding noise. The contour edges of the main change areas are effectively detected, which proves that, even in the unregistered image, SSALN can effectively utilize the labeled data to learn the changed difference information proactively. This excellent feature benefits from the low dependence of the model on true labels and the high reliability of the updated pseudolabels from a combinatorial way.

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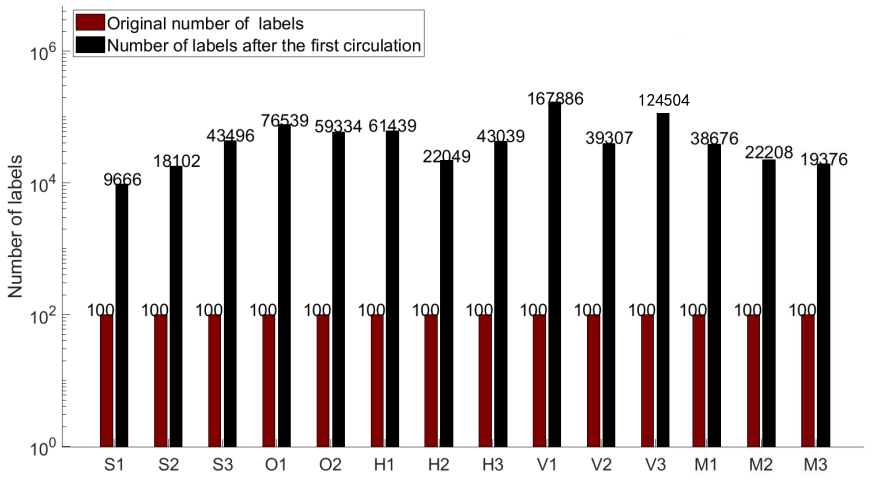


Fig. 21. Number of new pseudolabels added after the first circulation on each remote sensing image dataset. Note that S1 means the first dataset of SAR image, and O2 is corresponding to the second dataset of optical image. A similar meaning can be applied on H1, V1, M1, and so on.

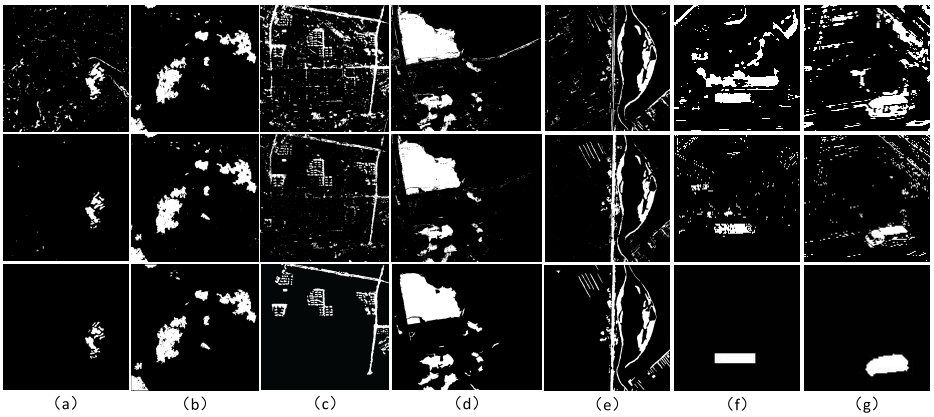


Fig. 22. First column and the second column are the CD results of the first and last circulation, respectively. The third column is the reference maps. (a) SAR image. (b) Optical image. (c) Heterogeneous image. (d) VHR image. (e) Multispectral image. (f) UAV1 image. (g) UAV2 image.

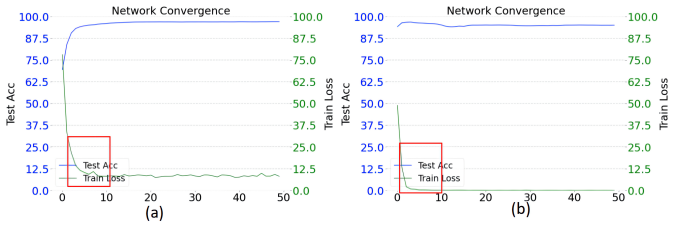


Fig. 23. Illustration of network convergence in (a) Bern dataset of the SAR image and (b) Mexico City dataset of the optical image.

E. Effect of Network Circulation

Overall, as the core of the framework, it is necessary to prove the validity and reliability of the circular part.

1) Convergence of Network in one Circulation: As shown in Fig. 23, the experiment is carried out on the first image pair, the Bern dataset, of SAR image in part (a) and Mexico City of the optical image in part (b) with 50 iterations, which is taken as an example. Here, we take the stability of the test accuracy as the standard and consider that the network has converged when the loss oscillation is less than a certain range. Basically, the network has converged within five iterations, which is shown in the red box in Fig. 23. The train loss and test accuracy have stabilized and met the demand.

Then, for network circulation, after achieving more reliable pseudolabels, the whole network should present an upward performance improvement. The problem is excepted to be analyzed from two perspectives in the following.

2) Number of Available Labels in Network Training: The pseudolabels of the whole framework are selected based on the classification prediction labels generated by the unsupervised method and the semisupervised network.

To test the extent to which reliable labels increase after the circulation, the number of obtained labels after the first circulation is compared with 100 initial real labels. Fig. 21 shows the impressive increase in the number of available labels. The amount of available labels has achieved a qualitative leap, which greatly enriches the information that can be processed by neural networks and proves that the framework can effectively use the potential information of unlabeled data. As can be seen, in the first dataset of the Bern dataset in SAR images, the original number of labels applied in

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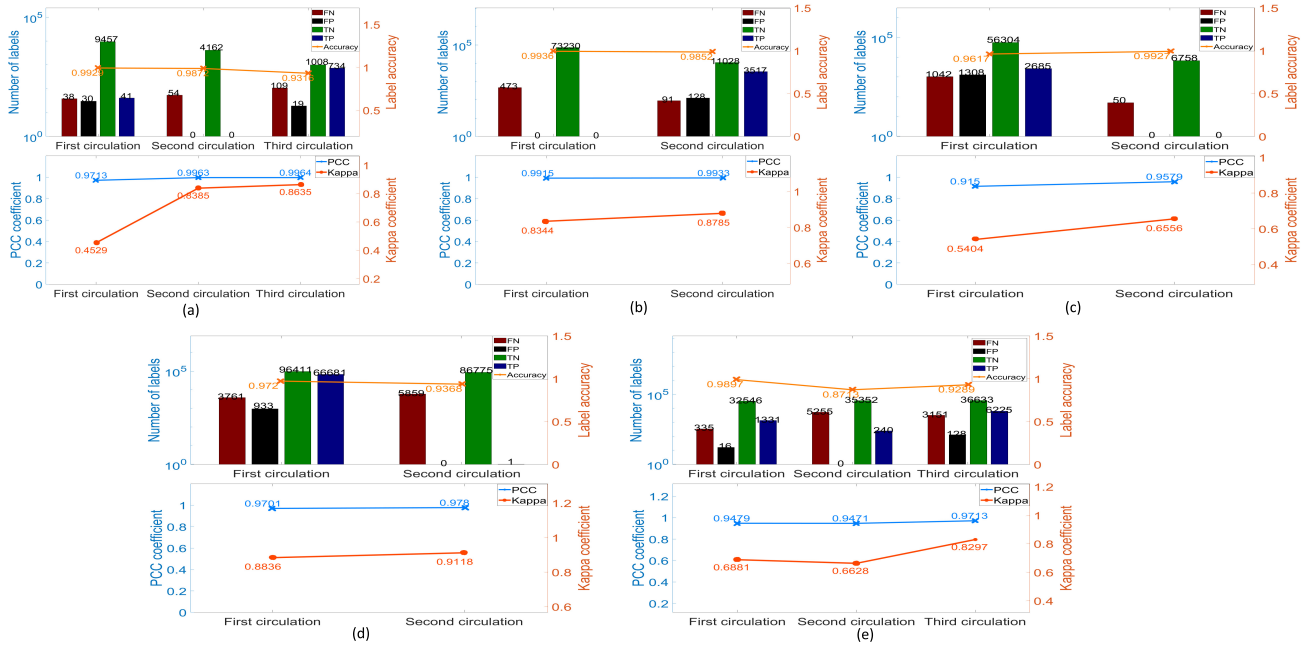


Fig. 24. Exhibition of the necessity of circulation on the increase in label number and accuracy and the coefficient of PCC and kappa based on the first set of different datasets. (a) SAR dataset. (b) Optical dataset. (c) Heterogeneous dataset. (d) VHR dataset. (e) Multispectral dataset.

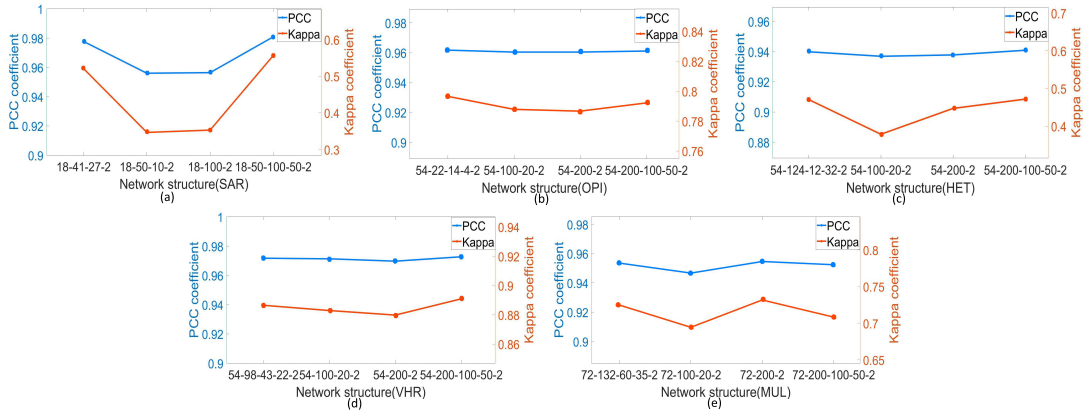


Fig. 25. Exhibition of the effective of SSALN on the several network structures based on each of the first set of different datasets. (a) SAR dataset. (b) Optical dataset. (c) Heterogeneous dataset. (d) VHR dataset. (e) Multispectral dataset.

network learning is 100, while, after the first circulation, the pseudolabels that can be utilized increase to 9666. Such a tremendous increase is also reflected in other images.

Fig. 22 clearly shows that the CD performance of the model finally increases with the circulation of the network. In the first circulation, the number of labels is 100, which is too small to detect the complex marginal areas and will produce a more serious unavoidable overfitting phenomenon, while, after iterations, with the increase in available labels, the detailed detection area gets better detection results in Column 2 of Fig. 22. The ability to suppress noise and maintain detail has been significantly improved.

3) Reliability of new Achieved Predictive Labels: The number of correct detected samples of the unchanged area and the changed area, and incorrect detected samples of the unchanged area and the changed area [TN, TP, false negative (FN), and false positive (FP)] in the predictive result and the accuracy with reference to ground-truth maps are shown in Column 1 of Fig. 24. It can be seen that TN and TP account for the majority of all predictive labels, and the quality of predictive labels is basically more than 90%, which means that most of them are reliable. Thus, those pseudolabels can be used to train neural networks circularly again to obtain better classification ability, which can be verified by high PCC and kappa coefficients in Column 2 of Fig. 24. Besides, when the label accuracy, PCC, and kappa coefficient are basically unchanged, the circulation will be stopped. Based on a large number of experiments, the network performance is stable after three circulations.

F. Effect of SSALN: Network Structure Adjustment

To prove the effectiveness of structure searching, the performance of autonomous-searched net and artificial-designed net

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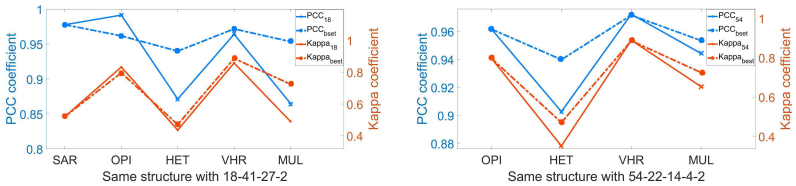


Fig. 26. Exhibition of the same net structure of SSALN on the different datasets. (a) Same network structure based on all of the first set of different datasets. (b) Same network structure based on datasets with multiple spectral bands.

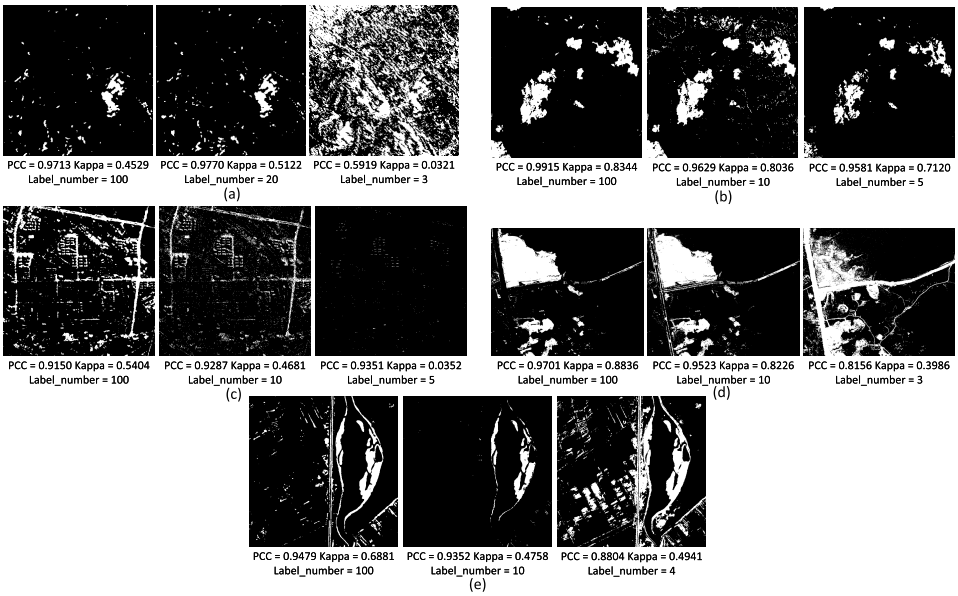


Fig. 27. Test of minimum number of true labels adopted in the network training based on the first set of different datasets. (a) SAR dataset. (b) Optical dataset. (c) Heterogeneous dataset. (d) VHR dataset. (e) Multispectral dataset.

on the first dataset in different types of remote sensing image is shown in Fig. 25.

As shown in Fig. 25(a)–(e), the performances in models of different network structures on the same image are different. The comparison of the better network structure searched by SSALN and other experienced networks proves the effectiveness of network architecture searching on a single image. For example, in the SAR image of Fig. 25(a), the performance of the network with the searched 18-41-27-2 structure is close to the performance of the manual-defined experienced network with the 18-50-100-50-2 structure, transcending the ordinary manual-defined structures of 18-50-10-2 and 18-100-2. Thus, the network performance of the structure searched by SSALN is not inferior to or even better than the one of the experienced artificial design networks. In addition, SSALN also solves the problem that the same network cannot intelligently learn different information proactively when detection objects are switched, which leads to the good performance of the same network in SAR datasets but worse performance in the heterogeneous image.

In order to explore the necessity of network structure adjustment during data switching, the experiment of the same searched net structure performing on different types of remote sensing image is carried out in Fig. 26.

Fig. 26(a) shows the network with the 18-41-27-2 structure that searched separately for SAR images performs well in the SAR image dataset but worse in other remote sensing image types. Because there is only one spectral segment in the SAR image, its processed dimension is not consistent with other types of images. Thus, on this basis, we measure the remote sensing image data types with the same number of spectral segments. The network structure of 54-22-14-4-2 that searched separately for the optical image is carried out on the heterogeneous image, VHR image, and multispectral image in Fig. 26(b). The results show that, although their input dimensions are consistent, the network that performs best in optical images does not achieve excellent results in other remote sensing image types, which indicates that the difference information of different remote sensing image datasets is complex. However, SSALN can learn the difference information proactively and complete the searching for the most suitable structure autonomously without human operation and achieve the best detection performance in a specific remote sensing image data type.

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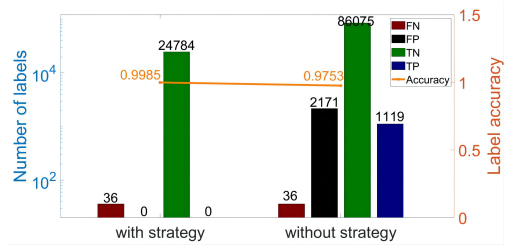


Fig. 28. Different results of pseudolabels in the SAR image.

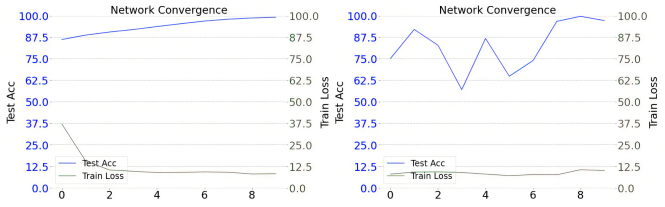


Fig. 29. Illustration of network convergence in the Bern dataset of the SAR image. (left: with-strategy result and right: without-strategy result).

G. Network Feasibility on Minimum Number of Labels

1) Effectiveness of the Presented Pseudolabel Updating Strategy: Before analyzing the minimum number of labels, we verify the effectiveness of our pseudotag update strategy.

1) *From the Accuracy of Pseudolabels:* First, the presented pseudolabel updating strategy is based on the semisupervised classification and unsupervised analysis. The unsupervised analysis is used to additionally help the proposed SSALN with pseudolabel updating from a completely different perspective. Thus, we particularly tested the coefficient of FN, FP, TN, TP, and the label accuracy of new pseudolabels without unsupervised analysis in Fig. 28.
   It clearly shows that, with the presented pseudolabel updating strategy, although the number of available labels has decreased, its accuracy has increased to a certain extent. It is worth noting that, in the without-strategy result, the number of FP is greater than TP. This means that the error part accounts for the majority in the area detected as change, which will greatly weaken the learning ability of network on the difference information between changed and unchanged areas.
2) *From the Convergence of Network Training:* For there being more wrong-divided detection points than correct-divided detection points in the changed area, the convergence of the without-strategy network is very difficult compared to with-strategy one, which is shown in Fig. 29. Although the learning rate is very small, the without-strategy network is also very difficult to train.
   Thus, the presented pseudolabel updating strategy ensures the accuracy of pseudolabels, suppressing the influence of wrong-divided samples and improving the stability of the network. It, therefore, forms the ascending network structure optimization.

2) Possible Minumum Number of Label Assumption: The pseudolabel updating strategy works depending on available labels. In the beginning, the proposed framework is carried out based on 100 true labels. In order to explore the greater practicability of the network, an experiment with decreasing number of adopted labels is carried out. It can be seen that the test results of minimum label amount are not consistent on different types of remote sensing images in Fig. 27. After more than ten times of retraining, a stable minimum number of labels used in the network can be obtained to avoid accidental existence. The final test shows that, even within ten labels, the network is basically steady. The classification accuracy of initial training is up to 90%, so the circulations can be still completed.

In fact, in addition to the complexity of the remote sensing image, the pixel size of the remote sensing image is an important index to measure the true labels needed by the proposed semisupervised network. It can be seen that the SAR image with few pixels can still complete the CD task under the label amount of 3 in Fig. 27(a), while others maybe not satisfactory even under the label amount of 100. In more tests, 0.5% of the total data of remote sensing image can basically meet the requirements of the CD task, which is shown in Table VII. SSALN under 0.5% labels achieves a much better performance. Thus, how to maintain the feature learning ability of the network with fewer labels is also in our future research.

TABLE VII
NEW COMPARED EVALUATION INDEXES OF THE HETEROGENEOUS IMAGE

Dataset	Methods	PCC	Kappa
Wuhan	SSALN(100)	0.9579	0.6556
	SSALN(0.5%)	**0.9722**	**0.7081**
Sardinia	SSALN(100)	0.9816	0.6877
	SSALN(0.5%)	**0.9820**	**0.7288**
Shuguang	SSALN(100)	0.9612	0.7764
	SSALN(0.5%)	**0.9656**	**0.8057**

V. CONCLUSION

In this article, an SSALN is proposed for remote sensing image CD. It is suitable for CD from two remote sensing images of any type with an iterative frame structure. After the sample selection and circulation of the network, the hidden information in labeled data completes the global diffusion in network training, which makes the high-order features extracted from unlabeled data synchronously conducive to classification without complex structure generation and complex structural analysis. Therefore, complex remote sensing image CD can be completed with only a small number of labels, greatly reducing the burden of manual marking. Besides, semisupervised and unsupervised CD approaches are combined together to guarantee the reliability of the predictive labels to support dynamic structural adjustment for incremental prior information. Multiple CD maps are fused to generate labels for the next iteration, and the disposable principle also prevents the incorrect-divided labels from polluting the network, improving the process of label prediction. Finally, both the fundamental and derivative analyses on five types

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of remote sensing images and UAV images have verified the universality and effectiveness of the SSALN and the ability to learn variable differences of multitype remote sensing images proactively. Even if the detection object is switched, it can be well adaptive and compatible without manual modification of the network.

However, the detection results of misaligned UAV images show that, although the SSALN performs well in the calibrated images, the performance of the images without registration will degrade seriously. This is due to the serious difficulty of label selection when the image is not calibrated, so how to maintain the ability of classification and promote the stability of the network in the case of lack of image alignment will form the central focus in our future work.

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